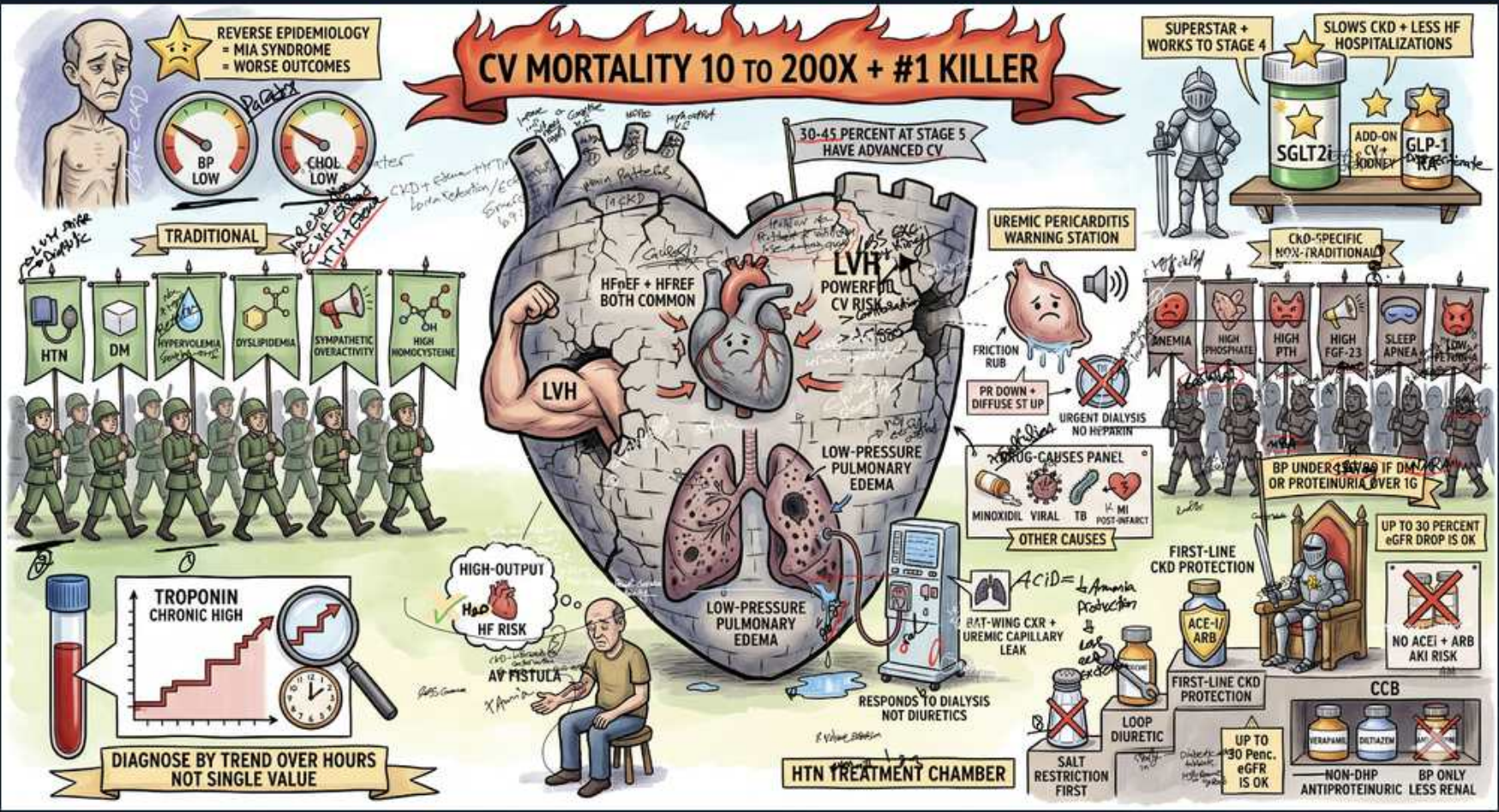


CKD — Cardiovascular Mortality (#1 Killer)



1. CV mortality 10–200x banner

WHAT YOU SEE

Banner 'CV mortality 10 to 200x + #1 killer'.

WHAT IT ENCODES

Cardiovascular disease is the #1 cause of death in CKD across all stages. CV risk rises as eGFR falls and albuminuria rises (each is an INDEPENDENT CV risk factor), with relative CV mortality up to ~10–20x normal in advanced CKD and reaching ~100–200x in young dialysis patients. CKD is considered a coronary-disease 'risk equivalent.' Mnemonic: in CKD, the heart kills before the kidney does.

BOARD TRAP

Stem: what is the most likely cause of death in a CKD patient, or what is the most common OUTCOME of CKD — wrong answer is 'progression to ESRD/dialysis' or 'uremia.' Discriminator: most CKD patients DIE of cardiovascular disease BEFORE ever reaching dialysis. CKD is far more likely to kill via CV events than to progress to renal replacement therapy.

2. Reverse epidemiology man

WHAT YOU SEE

Gaunt figure 'reverse epidemiology', BP low / chol low gauges, MIA syndrome.

WHAT IT ENCODES

In ADVANCED CKD/dialysis, classic risk relationships invert ('reverse epidemiology'): LOW BMI, LOW cholesterol, and LOW blood pressure predict WORSE survival, because they mark the Malnutrition-Inflammation-Atherosclerosis (MIA) syndrome / protein-energy wasting rather than cardiovascular health. Mnemonic: in dialysis, 'skinny/low numbers' = sick, not healthy.

BOARD TRAP

Stem: dialysis patient with low cholesterol, low BP, and weight loss is described as having 'good cardiovascular risk-factor profile' — wrong answer is to reassure or to aggressively lower numbers further. Discriminator: in advanced CKD these low values signal frailty/inflammation/malnutrition (MIA) and worse prognosis (reverse epidemiology) — they are NOT markers of low CV risk as they would be in the general population.

3. Traditional risk factors army

WHAT YOU SEE

Rows of soldiers 'traditional' HTN, DM, dyslipidemia, etc.

WHAT IT ENCODES

Traditional Framingham risk factors (hypertension, diabetes, dyslipidemia, smoking, age) are highly prevalent in CKD and contribute to CV disease, but they UNDER-predict the excess risk. Non-traditional/uremia-related factors fill the gap: vascular calcification (CKD-MBD), volume overload/LVH, anemia, inflammation, oxidative stress, hyperphosphatemia/FGF-23, and uremic toxins. Mnemonic: 'traditional soldiers aren't enough — uremia brings its own army.'

BOARD TRAP

Stem asks why a CKD patient with well-controlled BP, lipids, and glucose still has very high CV mortality — wrong answer is 'their traditional risk factors must be poorly controlled.' Discriminator: NON-traditional, CKD-specific factors (vascular calcification, LVH, FGF-23, anemia, inflammation, hyperphosphatemia) drive much of the excess risk beyond what traditional factors explain.

4. LVH muscle heart

WHAT YOU SEE

Muscular heart labeled 'LVH', 'HFpEF + HFrEF both common'.

WHAT IT ENCODES

Left ventricular hypertrophy is the MOST common cardiac structural abnormality in CKD, driven by pressure overload (hypertension, arterial stiffness), volume overload (salt/water retention, anemia, AV fistula), and uremic/FGF-23-mediated myocardial remodeling/fibrosis. It predisposes to both HFpEF (diastolic, more common) and HFrEF, arrhythmia, and sudden cardiac death. Mnemonic: the CKD heart is 'overworked muscle,' not just clogged arteries.

BOARD TRAP

Stem: CKD patient with dyspnea, normal coronaries, preserved EF, and LVH on echo — wrong answer is to attribute all CKD cardiac death to atherosclerotic CAD/MI. Discriminator: much CKD cardiac death is from LVH-related diastolic dysfunction, heart failure, arrhythmia, and SUDDEN CARDIAC DEATH (the single largest cardiac mode of death in dialysis), not solely coronary atherosclerosis.

5. Low-pressure pulmonary edema lungs

WHAT YOU SEE

Lungs labeled 'low-pressure pulmonary edema', responds to dialysis not diuretics.

WHAT IT ENCODES

Uremia increases pulmonary capillary permeability ('uremic lung'/leaky capillaries), so pulmonary edema can occur even at near-normal left-atrial filling pressures, on top of volume overload from impaired sodium/water excretion. It is fluid-overload-driven and clears with FLUID REMOVAL (ultrafiltration/dialysis). Mnemonic: 'wet lungs even when the pressures look dry' — clear it by pulling fluid off.

BOARD TRAP

Stem: anuric/oligo dialysis patient with flash pulmonary edema not responding to IV furosemide — wrong answer is to keep escalating loop diuretics. Discriminator: in advanced CKD/ESRD with little urine output, diuretics fail; the treatment is ULTRAFILTRATION/urgent dialysis to remove volume. Diuretics work only if the kidney can still respond.

6. Uremic pericarditis station

WHAT YOU SEE

'Uremic pericarditis warning station', friction rub, 'urgent dialysis'.

WHAT IT ENCODES

Uremic pericarditis presents with pleuritic chest pain and a pericardial FRICTION RUB; classically the ECG LACKS the diffuse ST elevation of typical viral pericarditis (because inflammation spares the myocardium). It signals severe uremia and carries risk of hemorrhagic effusion/tamponade. Mnemonic: 'rub + rising BUN = dialyze now.'

BOARD TRAP

Stem: uremic patient with chest pain and a friction rub — wrong answer is to treat with NSAIDs/colchicine as you would viral pericarditis. Discriminator: uremic pericarditis is an INDICATION FOR URGENT/INTENSIFIED DIALYSIS (part of the AEIOU dialysis indications); NSAIDs are avoided in CKD and don't address the cause, and anticoagulation/heparin should be minimized given hemorrhagic-effusion/tamponade risk.

7. Troponin chronic high / trend

WHAT YOU SEE

Tube + rising graph 'troponin chronic high', 'diagnose by trend over hours not single value'.

WHAT IT ENCODES

Troponin (especially high-sensitivity troponin T) is **CHRONICALLY** elevated at baseline in CKD due to reduced renal clearance, subclinical myocardial injury, and LVH — so a single elevated value is **NOT** diagnostic of acute MI. Diagnose ACS by a significant DELTA/rise-and-fall on **SERIAL** troponins over hours, integrated with symptoms and ECG. Mnemonic: in CKD read the troponin **TREND**, not the number.

BOARD TRAP

Stem: dialysis patient with vague symptoms and a single troponin above the reference limit — wrong answer is to diagnose MI off that one value (or to dismiss all elevations as 'just renal'). Discriminator: chronic baseline elevation is expected in CKD; you need a **SERIAL** troponin trend (significant delta) plus clinical/ECG correlation to call ACS. A flat, persistently elevated troponin favors chronic renal retention over acute MI.

8. High-output / AV fistula heart risk

WHAT YOU SEE

Man with arm, 'high-output', 'AV fistula HF risk'.

WHAT IT ENCODES

A large arteriovenous (dialysis) fistula shunts blood directly from artery to vein, increasing venous return and cardiac output; a high-flow fistula (especially >2 L/min or flow >20–30% of cardiac output) can cause **HIGH-OUTPUT** heart failure and aggravate LVH. Diagnosed by the Nicoladoni-Branham sign (heart rate slows with fistula compression). Mnemonic: the access that saves the kidney can overload the heart.

BOARD TRAP

Stem: dialysis patient with a high-flow brachial fistula develops heart failure with high cardiac output and warm extremities — wrong answer is to treat purely as low-output/fluid-overload HFrEF. Discriminator: consider **HIGH-OUTPUT** failure from the AV access (positive Branham sign, high-flow on access duplex); management may include flow-reducing fistula revision/banding, not just diuretics/afterload reduction.

9. SGLT2 add-on shelf

WHAT YOU SEE

Shelf 'SGLT2', 'slows CKD + less HF hospitalizations', GLP-1 add-on.

WHAT IT ENCODES

SGLT2 inhibitors (dapagliflozin, empagliflozin) reduce heart-failure hospitalizations and CV death **AND** slow CKD progression — benefits proven even in **NON-diabetic** proteinuric CKD (DAPA-CKD, EMPA-KIDNEY) and in HFpEF/HFrEF regardless of diabetes. They cause a small, expected, reversible early eGFR 'dip.' GLP-1 agonists add CV/renal/weight benefit. Mnemonic: SGLT2i = the cardio-renal cornerstone (protects heart **AND** kidney).

BOARD TRAP

Stem: non-diabetic CKD patient with proteinuria — wrong answer is to withhold an SGLT2 inhibitor because the patient 'isn't diabetic' or because eGFR dipped after starting it. Discriminator: SGLT2i benefit is **NOT** limited to diabetics (proven in non-diabetic CKD and HF), and the small early eGFR dip is hemodynamic and expected — not a reason to stop.

10. Statin — CKD-specific note

WHAT YOU SEE

Shelf 'CKD-specific vs traditional' statin note.

WHAT IT ENCODES

Statins (often statin + ezetimibe, per SHARP) reduce atherosclerotic CV events in **NON-dialysis** CKD and are recommended for most CKD patients ≥50 (and selected younger ones). **BUT** trials of **INITIATING** statins in **DIALYSIS** patients (4D in diabetics on HD, AURORA) showed **NO** clear reduction in CV mortality — reflecting non-atherosclerotic mechanisms of death (sudden cardiac death, HF, calcification). Mnemonic: 'statins help the kidney patient — until they start dialysis.'

BOARD TRAP

Stem: stable hemodialysis patient with no prior statin is started on atorvastatin for primary CV prevention — wrong answer is that this clearly reduces mortality. Discriminator: do **NOT** initiate statins de novo for primary prevention in **DIALYSIS** patients (4D/AURORA showed no benefit); however, patients already on a statin when they start dialysis are generally continued, and non-dialysis CKD patients **DO** benefit (SHARP).

11. ACEi/ARB first-line knight

WHAT YOU SEE

Knight 'first-line CKD protection — ACE/ARB', 'up to 30% creatinine OK'.

WHAT IT ENCODES

ACE inhibitors/ARBs are first-line in CKD for BP control **AND** renoprotection, especially with albuminuria/proteinuria — they dilate the efferent arteriole, lowering intraglomerular pressure and proteinuria. Expect (and tolerate) an early creatinine rise up to ~30% and a small potassium rise. Mnemonic: a little creatinine bump means the drug is doing its job (lowering glomerular pressure).

BOARD TRAP

Stem: proteinuric CKD patient started on an ACEi has creatinine rise 20% and potassium 5.1 — wrong answer is to stop the drug immediately. Discriminator: a creatinine rise up to ~30% (and mild hyperkalemia) is **EXPECTED** and acceptable — continue and monitor; stop/work up only if the rise exceeds ~30% (suspect bilateral renal artery stenosis/volume depletion). Also: do **NOT** combine ACEi + ARB (more AKI/hyperkalemia, no outcome benefit — ONTARGET).

12. CCB add-on

WHAT YOU SEE

Bottles 'CCB', non-DHP vs DHP note.

WHAT IT ENCODES

Calcium channel blockers are effective add-on antihypertensives in CKD. Dihydropyridines (amlodipine) are potent BP-lowering agents but can **WORSEN** proteinuria if used without RAAS blockade (afferent vasodilation raises glomerular pressure). Non-dihydropyridines (diltiazem, verapamil) additionally **REDUCE** proteinuria. Mnemonic: 'non-DHP also helps the protein leak.'

BOARD TRAP

Stem: proteinuric CKD patient with uncontrolled BP and an antiproteinuric goal — wrong answer is to add a dihydropyridine (amlodipine) **WITHOUT** a RAAS blocker on board. Discriminator: dihydropyridine CCBs lower BP but can increase intraglomerular pressure/proteinuria when used alone; pair them with an ACEi/ARB, or choose a **NON-dihydropyridine** (diltiazem/verapamil) when an antiproteinuric add-on is the goal.

13. Salt restriction chamber

WHAT YOU SEE

Caption 'salt restriction', 'HTN treatment chamber'.

WHAT IT ENCODES

Dietary sodium restriction (target ~<2 g sodium/day) is foundational in CKD: it lowers BP, reduces volume/edema, and crucially POTENTIATES the antiproteinuric and antihypertensive effect of ACEi/ARB and diuretics. A high-salt diet blunts RAAS-blocker efficacy. Mnemonic: 'salt sabotages your ACE inhibitor.'

BOARD TRAP

Stem: CKD patient with persistent proteinuria/HTN despite a maximized ACEi, eating a high-sodium diet — wrong answer is to immediately add a third agent before addressing diet. Discriminator: a high-salt intake BLUNTS RAAS-blocker effect; sodium restriction (and/or a diuretic) restores efficacy and should be optimized before escalating pharmacotherapy.

14. Loop diuretic

WHAT YOU SEE

Bottle 'loop diuretic'.

WHAT IT ENCODES

Loop diuretics (furosemide, torsemide, bumetanide) are the diuretics of choice for volume/edema and volume-dependent hypertension as GFR declines, because they still work at low eGFR (often needing HIGHER doses to reach the tubular site of action). They also help manage hyperkalemia and metabolic load. Mnemonic: 'loops keep working when the GFR is low.'

BOARD TRAP

Stem: volume-overloaded CKD patient at eGFR 25 prescribed hydrochlorothiazide for edema with poor response — wrong answer is to keep relying on the thiazide. Discriminator: thiazides lose efficacy as monotherapy at low GFR (roughly eGFR <30); switch to a LOOP diuretic for volume control in advanced CKD (a thiazide may be ADDED to a loop for sequential nephron blockade in diuretic resistance, but not used alone).